

Coding & Solution

Date	08 July 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/poshithayagna2007-max/VBCUA
Programming Language(s)	Python
Framework(s) Used	Streamlit
Key Features Implemented	• Voice recording and audio upload • Speech-to-Text using OpenAI Whisper • Topic selection (predefined) • Semantic similarity evaluation using Sentence-BERT • Audio feature extraction for fluency and clarity • AI-based conceptual understanding score • Personalized feedback generation • Downloadable PDF evaluation report • Interactive Streamlit dashboard
Pending / Incomplete Features	• Database integration for storing user history (optional) • Multi-language speech evaluation (future enhancement) • Advanced analytics dashboard (future enhancement)
Setup / Run Instructions	1. Clone the GitHub repository. 2. Create and activate a Python virtual environment. 3. Install dependencies using <code>pip install -r requirements.txt</code>. 4. Run the application using <code>streamlit run app.py</code>. 5. Open the displayed local URL in a web browser to access the application.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture, separating the user interface, speech processing, semantic evaluation, audio analysis, and report generation into independent components. The code adheres to Python coding standards, making it readable, maintainable, and reusable. The application successfully demonstrates AI-powered evaluation of conceptual understanding through voice responses and provides detailed performance feedback.